

[Figure 7 — placeholder]

Six-panel grid showing the optimization trajectory for the explicit inverse scheme on **Sample 1** (the homogeneous case, target SM = 3.00 MPa), starting from a 6-layer initial guess. Each panel is a rectangular shear-modulus distribution (1.0 mm × 0.2 mm) with a jet colorbar.

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| (a) Initial guess | uniform blue, SM = 1.00 MPa everywhere |
| (b) Iteration 5 | SM range 1.27–1.95, layered artifacts emerging |
| (c) Iteration 12 | SM range 1.50–3.11, layered structure developed |
| (d) Iteration 24 | SM range 1.63–3.03, approaching target |
| (e) Iteration 48 | SM range 2.97–3.00, nearly converged |
| (f) Iteration 52 (Final) | uniform red, SM = 3.00 MPa everywhere |

The trajectory shows the 6-layer parameterisation collapsing to a uniform solution over 52 iterations — because the target is homogeneous, the optimiser correctly drives all six layers to the same shear modulus.

Fig. 7. The optimization process with six layers initially utilizing the explicit inverse scheme for Sample 1 (Without noise, Unit: MPa).

[Figure 8 — placeholder]

Six-panel side-by-side comparison of the **Explicit Inverse Method** (left column) vs. **Implicit Inverse Method** (right column) on Sample 1 (homogeneous, target SM = 3.00 MPa), with 3, 6, and 9 surface displacement datasets as inputs.

Explicit (left column): all three rows show a perfectly uniform shear-modulus field at SM = 3.00 MPa (uniform blue, colorbar pinned 3.00–3.00). The explicit parameterisation recovers the homogeneous target exactly.

Implicit (right column): all three rows show tiny spatial artifacts around the true value of 3.00 MPa — (d) 3 fields: SM range 2.9995–3.0001; (e) 6 fields: SM range 2.9992–3.0001; (f) 9 fields: SM range 3.0000–3.0002. The color blobs are visible even though the numerical error is on the order of 10^{-4} – 10^{-3} MPa.

Takeaway: the explicit method produces a perfectly uniform field on the homogeneous case; the implicit method is quantitatively accurate but shows residual spatial noise patterns that decrease (slightly) as more displacement datasets are incorporated.

Fig. 8. The reconstructed results with varying 3, 6 and 9 surface displacement datasets corresponding to Sample 1 (Without noise, Unit: MPa).

inverse method shows a slight spatial variation, which can be enhanced by incorporating more surface datasets.

After adding 1 % noise to the surface displacement datasets, we observed that the shear modulus distributions were accurately mapped for both inverse methods in the homogeneous case (refer to Fig. 11). The average relative error for both implicit and explicit inverse methods was less than 0.2 % (see Table 1). However, when it came to Sample 2 and 3 (see Fig. 12, Fig. 13), the interface between neighboring layers in the implicit inverse method's results was not as clear as the target. As the total number of layers increases, the challenges in solving the inverse problems escalate accordingly. Consequently, the relative error for Sample 2 and Sample 3 exhibits a lower quality compared to Sample 1 (see Table 1). Besides, the average relative error for the implicit inverse method was